

JEE-Mains-08-04-2023 [Memory Based] [Morning Shift]

Physics

Question: Body weighs 400N on surface of earth find the weight of the body at a depth $R/2$ from the surface of earth (R = radius of earth)

Options:

- (a) 100 N
- (b) 200 N
- (c) 300 N
- (d) 400 N

Answer: (b)

Solution:

$$F_1 = \frac{400N}{F_2 N} = \frac{mgs}{mgs \left(1 - \frac{d}{R}\right)}$$

where $d = R/2$

$$\frac{400}{F_2} = \frac{1}{\left(1 - \frac{1}{2}\right)}$$

$$\Rightarrow F_2 = 200$$

Question: If $m = 1/2$ kg, $\vec{v} = (2t)\hat{i} + (3t^2)\hat{j}$. If at $t = 1$ sec $\vec{F} = \hat{i} + x\hat{j}$. Find x?

Answer: 3.00

Solution:

$$\vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt}$$

$$= \frac{1}{2} \left[\frac{d}{dt} (2t\hat{i} + 3t^2\hat{j}) \right]$$

$$= \frac{1}{2} (2\hat{i} + 6t\hat{j})$$

$$\vec{F} = \hat{i} + 3t\hat{j} \text{ at } t = 1 \Rightarrow F = \hat{i} + 3\hat{j}$$

So, $x = 3$

Question: If momentum of body is increased by 50% find percentage increase in Kinetic Energy

Options:

- (a) 80%
- (b) 100%
- (c) 50%
- (d) 125%

Answer: (d)

Question: Statement 1: heat is given to a gas, its temp must increase

Statement 2: if positive work is done , volume must increase

Options:

- (a) S1 - False, S2 - True
- (b) S1 - True, S2 - True
- (c) S1 - True, S2 - False
- (d) S1 - False, S2 - False

Answer: (a)

Question: Two forces of magnitude A and $\frac{A}{2}$ act perpendicular to each other. The magnitude of the resultant force is equal to

Options:

- (a) $A/2$
- (b) $\sqrt{5}A/2$
- (c) $3A/2$
- (d) $5A/2$

Answer: (b)

Solution: $\sqrt{A^2 + \frac{A^2}{4}} = \frac{\sqrt{5}}{2} A$

Question: An antenna has height 98m find distance upto which signal can be transmitted (R = 6400 km)

Options:

- (a) 3642 km²
- (b) 3942 km²
- (c) 11200 km²
- (d) 22400 km²

Answer: (b)

Solution:

Question: Two projectiles of speed $u_1 = 40 \text{ m/s}$ and $u_2 = 60 \text{ m/s}$ placed at a angle of 30° and 60° respectively. Find the ratio of range.

Answer:

Solution:

$$R_1 = \frac{40^2 \sin 60^\circ}{g}$$

$$R_2 = \frac{60^2 \sin 120^\circ}{g}$$

$$\frac{R_1}{R_2} = \left(\frac{40}{60} \right)^2 = \frac{4}{9}$$

Question: An air bubble having volume 1 cm³ at depth 40 m inside water comes to surface. What will be the volume of the bubble at the surface.

Options:

- (a) 5 cm^3
- (b) 2 cm^3
- (c) 4 cm^3
- (d) 3 cm^3

Answer: (a)

Solution:

Assuming isothermal condition,

$$P_1 V_1 = P_2 V_2$$

$$(P_0 + \rho gh) V_0 = P_0 V_f$$

$$[10^5 + 10^3 \times 10 \times 40] 1 \text{ cm}^3 = 10^5 V_f$$

$$5 \times 10^5 \times 1 \text{ cm}^3 = 10^5 V_f$$

$$V_f = 5 \text{ cm}^3$$

Question: For an electron and a proton ($m_p = 1847 m_e$) with same de-Broglie wavelength, the ratio of linear momentum is equal to:

Options:

- (a) 1 : 2
- (b) 2 : 1847
- (c) 1 : 1
- (d) $\sqrt{1847} : 1$

Answer: (a)

Question: If $Y = 7 \times 10^{11} \text{ Nm}^{-1}$ & $\Delta L/L = 0.04$. Find energy density.

Solution:

$$\left(\frac{U}{V} \right) = \frac{1}{2} Y \epsilon^2$$

$$= \frac{1}{2} \times 7 \times 10^{11} \times (0.04)^2$$

$$= \frac{16 \times 10^{-4} \times 7 \times 10^{11}}{2} = 8 \times 7 \times 10^{-7}$$

$$= 5.6 \times 10^{-6} \text{ (J/m}^3 \text{)}$$

Question: The height of antenna is 98 m. The radius of Earth is 6400 km. The area up to which it will transmit signal is:

Options:

- (a) 3642 km^2
- (b) 3942 km^2
- (c) 11200 km^2
- (d) 22400 km^2

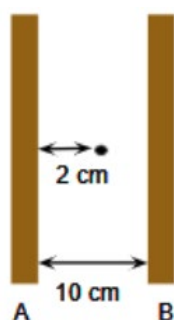
Answer: (b)

Solution:

Area

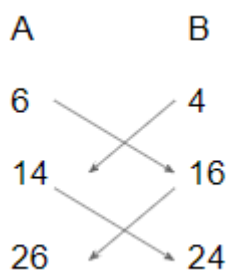
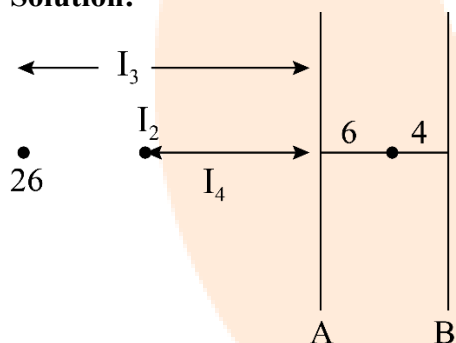
$$\begin{aligned}
 &= \pi (\text{Range})^2 \\
 &= \pi \left[\sqrt{2hR_E} \right]^2 \\
 &= 2\pi hR_E \\
 &= 2 \times 3.14 \times 98 \times 6400 \times 10^3 \\
 &\approx 3940 \times 10^6 \text{ (m}^2\text{)} \\
 &\text{or } 3942 \text{ (km}^2\text{)}
 \end{aligned}$$

Question: In the given diagram, find the distance between 2nd and 3rd image formed left of mirror A.



Answer: 12.00

Solution:



$$\text{Gap} = 26 - 14 = 12$$

Question: If mass radius of cross-section and height of a cylinder are (0.4 ± 0.01) g, (6 ± 0.03) m and (8 ± 0.04) m. The maximum percentage of error in the measurement of density of cylinder is:

Options:

- (a) 1%
- (b) 4%

(c) 8%

(d) 7%

Answer: (b)

Solution:

$$\rho = \frac{M}{\pi r^2} \cdot h$$

$$\Rightarrow \frac{\Delta \rho}{\rho} \times 100 = \frac{\Delta M}{M} \times 100 + 2 \frac{\Delta r}{r} \times 100 + \frac{\Delta h}{h} \times 100$$

$$= \frac{0.01}{0.4} \times 100 + 2 \times \frac{0.03}{6} \times 100 + \frac{0.04}{8} \times 100$$

$$= 2.5 + 1 + 0.5 = 4\%$$

Question: If velocity of charged particle has the component both in and perpendicular to the direction of magnetic field then the path traced by the charged particle will be

Options:

(a) Circular

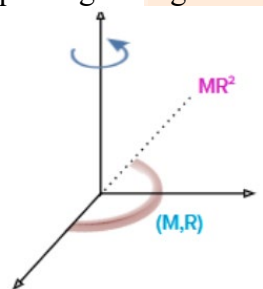
(b) Straight line

(c) Cycloid

(d) Helical

Answer: (d)

Question: The moment of inertia of semi-circular ring of mass m and radius R about an axis passing through centre and perpendicular to the plane of ring is



Options:

(a) MR^2

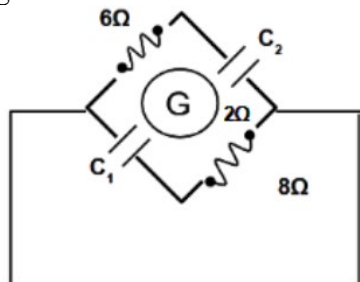
(b) $\frac{1}{2} MR^2$

(c) $2MR^2$

(d) $\frac{3}{4} MR^2$

Answer: (a)

Question: What is the ratio of potential difference across C_1 and C_2 at steady state for the given circuit

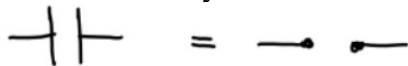


Options:

- (a) 4 : 5
- (b) 2 : 5
- (c) 1 : 4
- (d) 3 : 1

Answer: (a)

Solution: Steady State



$$\text{So, } i = \frac{4}{6+2+8} = \frac{1}{4}$$

$$\text{So } V_B = 4 - iR$$

$$= 4 - \frac{1}{4} \times 6$$

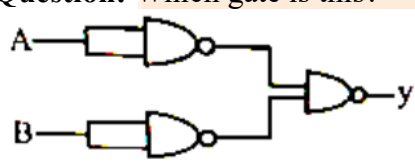
$$V_B = 2.5$$

$$V_C = V_B - iR$$

$$= 2.5 - \frac{1}{4} \times 2 = 2$$

$$\frac{V_{AC}}{V_{BD}} = \frac{4-2}{2.5-0} = \frac{2}{\frac{5}{2}} = \frac{4}{5}$$

Question: Which gate is this?



Options:

- (a) AND
- (b) OR
- (c) NOT
- (d) NOR

Answer: (b)

Question: In an LC oscillating circuit with $L = 75 \text{ mH}$ and $C = 30 \mu\text{F}$, the maximum charge of capacitor is $2.7 \times 10^{-4} \text{ C}$. Maximum current through the circuit will be

Options:

- (a) 0.18 Amp
- (b) 0.24 Amp
- (c) 0.72 Amp
- (d) 0.92 Amp

Answer: (a)

Solution:

$$i_{\max} = \frac{Q_{\max}}{\sqrt{LC}} = \frac{2.7 \times 10^{-4} \text{ C}}{\sqrt{75 \times 10^{-3} \times 30 \times 10^{-6}}}$$

$$= \frac{2.7 \times 10^{-4}}{15 \times 10^{-4}} = 0.18$$



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Chemistry

Question: The correct order of electronegativity B, S, C and At

Options:

- (a) $B > C > S > At$
- (b) $S > C > B > At$
- (c) $C > B > S > At$
- (d) $S > C > At > B$

Answer: (d)

Solution: Fact based

Question: Match the following:

Parameter	Maximum Prescribed conc.
(P)Fluorine	(1) 50 ppb
(Q)Lead	(2) 0.2 ppm
(R)Fe	(3) 1 ppm
(S)Nitrate	(4) 50 ppm

Options:

- (a) P-3, Q-1, R-2, S-4
- (b) P-3, Q-2, R-1, S-4
- (c) P-2, Q-1, R-3, S-4
- (d) P-1, Q-2, R-4, S-3

Answer: (a)

Solution: Fact based

Maximum Prescribed Concentration of Some Metals in Drinking Water.

Metal	Maximum concentration (ppm or mg dm^{-3})
Fe	0.2
Mn	0.05
Al	0.2
Cu	3.0
Zn	5.0
Cd	0.005

Question: Assertion: Butanol has highest boiling point than ethoxyethane.

Reason: Because butanol has more hydrogen bonding.

Options:

- (a) Both assertion and reason are correct and reason is correct explanation
- (b) Both assertion and reason are correct and reason is incorrect explanation
- (c) Both assertion and reason are incorrect
- (d) Assertion is correct but reason is incorrect

Answer: (a)

Solution: fact based

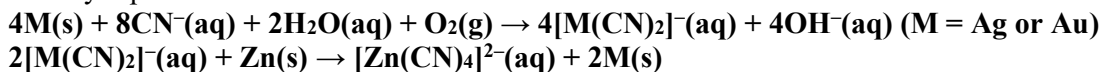
Question: Which element is purified by leaching

Options:

- (a) Pb
- (b) Sn
- (c) Cu
- (d) Au

Answer: (d)

Solution: In the metallurgy of silver and gold, the respective metal is leached with a dilute solution of NaCN or KCN in the presence of air, which supplies O_2 . The metal is obtained later by replacement reaction.



Question: $2Cu^{+2} + 4X_2 \rightarrow Cu_2X_2 + X_2$

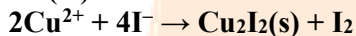
What is X here?

Options:

- (a) Iodine
- (b) Bromine
- (c) Chlorine
- (d) All of above

Answer: (a)

Solution: $2Cu^{+2} + 4X_2 \rightarrow Cu_2X_2 + X_2$ and the same applies to CuX . On the other hand, all $Cu(II)$ halides are known except the iodide. In this case, Cu^{2+} oxidises I^- to I_2 :

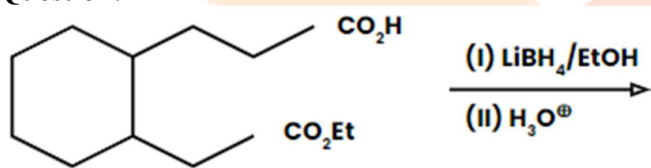


However, many copper (I) compounds are unstable in aqueous solution and undergo disproportionation.



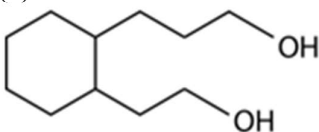
The stability of $Cu^{2+}(aq)$ rather than $Cu^+(aq)$ is due to the much more negative $\Delta_{hyd}H^\ominus$ of $Cu^{2+}(aq)$ than Cu^+ , which more than compensates for the second ionization enthalpy of Cu.

Question:

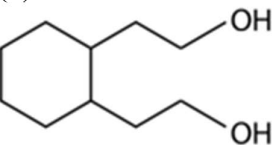


Options:

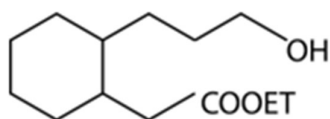
(a)



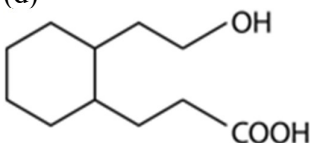
(b)



(c)

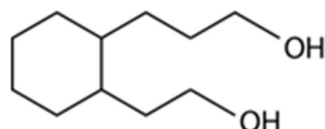


(d)



Answer: (a)

Solution:



Question: The chromium in chromyl chloride has the same oxidation state as which of the following

Options:

(a) Fe^{3+}

(b) V^{4+}

(c) Ti^{+2}

(d) Mn^{+6}

Answer: (d)

Solution: Chromyl chloride reaction is not redox reaction

Question: Match the following:

Column I	Column II
(P) Saccharine	(1) Sweetest sugar
(Q) Alitame	(2) Unstable on cooking
(R) Aspartame	(3) Stable at cooking temperature
(S) Sucralose	(4) First artificial sweetner

Options:

(a) P – 4; Q – 1; R – 2; S – 3

(b) P – 4; Q – 3; R – 2; S – 1

(c) P – 1; Q – 2; R – 3; S – 4

(d) P – 2; Q – 3; R – 4; S – 1

Answer: (a)

Solution: Some other commonly marketed artificial sweeteners are given in table.

Saccharin	First Artificial Sweetener
Alitame	More stable than aspartame
Aspartame	Unstable at cooking temperature
Sucralose	Stable at Cooking temperature

Question: How many statements are correct ?

Statement-1 : Li^+ has higher polarizing power than Mg^{2+}

Statement-2: Mg^{2+} ion is smaller than Li^{2+} ion.

Options:

(a) statement 1 is correct and statement 2 is incorrect

- (b) statement 1 is incorrect and statement 2 is correct
- (c) statement 1 is incorrect and statement 2 is also incorrect
- (d) statement 1 is correct and statement 2 is also correct

Answer: (b)

Solution: Mg has more polarizing power than Li,

Question: Which of the following contain Sulphur

Lysine, Methionine, glutamic acid , threonine, Arginine, Cystine, tyrosine, isoleucine

Options:

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (b)

Solution: Methionine and Cystine

Question: What is formed when water gas reacts in presence of Copper as catalyst

Options:

- (a) Ethanol
- (b) Methanal
- (c) Methanoic acid
- (d) Ethanoic acid

Answer: (b)

Solution: fact based

Question: How many of the following factors results in covalent character in a compound?

Polarising power of cation.

Polarisability of anion

Electron cloud distortion of anion

Electron cloud distortion of Cation

Options:

- (a) 3
- (b) 4
- (c) 2
- (d) 1

Answer: (a)

Solution: Polarisability of anion causes cloud distortion and polarizing power of cation causes covalent nature. FAJAN RULE

Question: Which Cell representation is correct for



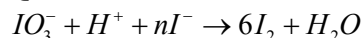
Options:

- (a) $\text{Pt} | \text{H}_2 | \text{HCl} | \text{AgCl} | \text{Ag}$
- (b) $\text{Pt} | \text{H}_2 | \text{HCl} | \text{AgCl} | \text{Pt}$
- (c) $\text{Ag} | \text{AgCl} | \text{HCl} | \text{H}_2 | \text{Pt}$
- (d) $\text{Pt} | \text{AgCl} | \text{HCl} | \text{H}_2 | \text{Pt}$

Answer: (a)

Solution: Fact based

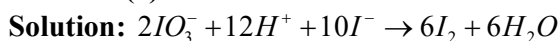
Question: Find the value of 'n' in the following redox reaction.



Options:

- (a) 10
- (b) 12
- (c) 9
- (d) 5

Answer: (a)



Question: For an electron and proton of same de-Broglie wavelength, the ratio of linear momentum is:

Options:

- (a) 1 : 2
- (b) 2 : 1847
- (c) 1 : 1
- (d) $\sqrt{1847} : 1$

Answer: (c)

Solution: calculation based $\lambda = \frac{h}{p}$

Question: How many statements are correct:

Options:

- (a) If there is no relation between rate constant and temperature, then activation energy is negative
- (b) If the activation energy is zero, rate constant is temperature independent
- (c) IF rate constant increases with increase of temperature, activation energy is positive
- (d) IF rate constant decreases with increase in temperature, activation energy is negative.

Answer: (c)

Solution: fact based

Question: Which of the following is most stable, diamagnetic and octahedral shaped:

Options:

- (a) $K_3[Co(CN)_6]$
- (b) $[Co(H_2O)_6]Cl_3$
- (c) $Na_3[CoF_6]$
- (d) All have exact equal stability

Answer: (a)

Solution: Hybridization is d^2sp^3

Question: For the ions: $[MnF_6]^{4-}$, $[Fe(CN)_6]^{3-}$ and $[Co(NH_3)_6]^{3+}$. The order of the spin magnetic moment is correct in which of the following option.

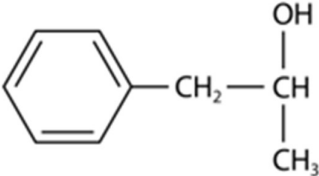
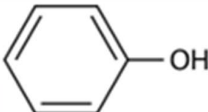
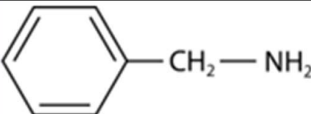
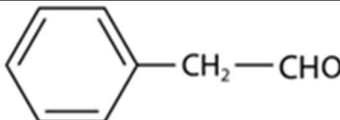
Options:

- (a) $[MnF_6]^{4-} > [Co(NH_3)_6]^{3+} > [Fe(CN)_6]^{3-}$
- (b) $[Fe(CN)_6]^{3-} > [MnF_6]^{4-} > [Co(NH_3)_6]^{3+}$
- (c) $[MnF_6]^{4-} > [Fe(CN)_6]^{3-} > [Co(NH_3)_6]^{3+}$
- (d) $[Co(NH_3)_6]^{3+} > [Fe(CN)_6]^{3-} > [MnF_6]^{4-}$

Answer: (c)

Solution:

Question: Match the column

Column I		Column II	
(a)	Neutral FeCl_3	(p)	
(b)	Iodoform	(q)	
(c)	Carbylamine test	(r)	
(d)	$[\text{CuSO}_4 + \text{Sodim potassium tartarate (Rochele's salt)}]$	(s)	

Options:

(a) A – Q; B – R; C – P; D – S

(b) A – P; B – R; C – Q; D – S

(c) A – Q; B – P; C – R; D – S

(d) A – Q; B – P; C – S; D – R

Answer: (c)

Solution:

Question: 0.5 g of an organic compound with 60% carbon will produce ___g of CO_2 upon complete combustion?

Options:

(a) 11 g

(b) 12 g

(c) 15 g

(d) 20 g

Answer: (a)

Solution: $0.5 \times 0.6 = 0.3 \text{ g of carbon}$

Moles of carbon $= \frac{0.3}{12} = 0.025$

$\therefore \text{CO}_2 = 0.025$

$wt = 0.025 \times 44 = 11\text{g}$

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Mathematics

Question: Find the maximum value of n such that $66!$ is divisible by 3^n .

Answer: 31.00

Solution:

Exponent of 3 in $66!$

$$\left[\frac{66}{3} \right] + \left[\frac{66}{3^2} \right] + \left[\frac{66}{3^3} \right] + \left[\frac{66}{3^4} \right]$$

$$\left[\frac{66}{3} \right] + \left[\frac{66}{3^2} \right] + \left[\frac{66}{3^3} \right] + 0$$

$$= 22 + 7 + 2$$

$$= 31$$

Question: Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & 1 & 2 \end{bmatrix}$ and $|\text{adj}(\text{adj}(\text{adj } 2A))| = 16^n$. Find n .

Answer: 12.00

Solution:

$$\text{Given } A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$|A| = 2 \times 5 - 1 \times 2 = 8$$

$$|\text{adj}(\text{adj}(\text{adj } 2A))| = |2A|^{2^3} = |2A|^8$$

$$= |2^3 A|^8 = (8^2)^8 = 8^{16}$$

$$= 2^{48} = (2^4)^{12} = 16^{12}$$

$$\Rightarrow n = 12$$

Question: $x, y, 12, 12, 10, 4, 8, 6$; Variance = 9.25 & mean = 9. If $x > y$, then find $3x - 2y$.

Answer: 25.00

Solution:

$$\frac{x + y + 52}{8} = 9$$

$$x + y = 20$$

$$x - 9, y - 9, 3, 3, 1, -5, -1, -3$$

$$\bar{x} = 0$$

$$\text{Now, Variance, } \frac{(x-9)^2 + (y-9)^2 + 54}{8} - 0^2 = 9.25$$

$$(x-9)^2 + (y-9)^2 = 20$$

$$(x-9)^2 + (11-x)^2 = 20$$

$$x = 7, 13$$

$$y = 13, 7$$

$$3x - 2y = 3 \times 13 - 2 \times 7 = 25$$

Question: Find the fourth term in the expansion of $(1+x)^n$, if the coefficient of three consecutive terms are in the ratio 1:5:20.

Answer: ${}^{29}C_3 x^3$

Solution:

$${}^nC_{r-1} = 1, {}^nC_r = 5, {}^nC_{r+1} = 20$$

$$\frac{{}^nC_r}{{}^nC_{r-1}} = 5 \Rightarrow \frac{n-r+1}{r} = 5 \Rightarrow n = 6r - 1$$

$$\frac{{}^nC_{r+1}}{{}^nC_r} = 4 \Rightarrow \frac{n-(r+1)+1}{r+1} = 4 \Rightarrow n = 5r + 4$$

$$\Rightarrow 6r - 1 = 5r + 4$$

$$\Rightarrow r = 5$$

$$\Rightarrow n = 29$$

$$t_4 = {}^nC_3 x^{n-3} = {}^{29}C_3 x^3$$

Question: $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $Q = PAP^T$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, then $P^T Q^{2007} P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Find

$$2a + b + 3c - 4d.$$

Answer: 2005.00

Solution:

$$\text{Given, } P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$PP' = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} = I$$

$$\Rightarrow P' = P^{-1}$$

$$Q = PAP^{-1}$$

$$\Rightarrow Q^{2007} = PA^{2007}P^{-1}$$

$$P'Q^{2007}P = P^{-1}PA^{2007}P^{-1}P = A^{2007}$$

$$\text{Now, } A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = I + B$$

$$B^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow B^n = 0 \forall n \geq 2$$

$$\text{So, } A^{2007} = (I + B)^{2007} = I + 2007B$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2007 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix}$$

$$2a + b + 3c - 4d = 2 + 2007 + 0 - 4 = 2005$$

Question: $f(x) = \frac{\sin x + \cos x - \sqrt{2}}{\sin x - \cos x}$. Find $f\left(\frac{\pi}{12}\right) + f''\left(\frac{\pi}{12}\right)$.

Answer: $30 - 17\sqrt{3}$

Solution:

$$f(x) = \frac{\sqrt{2} \left(\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x \right) - \sqrt{2}}{\sqrt{2} \left(\frac{1}{\sqrt{2}} \sin x - \frac{1}{\sqrt{2}} \cos x \right)}$$

$$f(x) = \frac{-\sqrt{2} \left[1 - \cos \left(x - \frac{\pi}{4} \right) \right]}{\sqrt{2} \sin \left(x - \frac{\pi}{4} \right)}$$

$$f(x) = -\tan \left(\frac{x}{2} - \frac{\pi}{8} \right)$$

$$f'(x) = -\frac{1}{2} \sec^2 \left(\frac{x}{2} - \frac{\pi}{8} \right)$$

$$f''(x) = \frac{-1}{4} \times 2 \sec^2 \left(\frac{x}{2} - \frac{\pi}{8} \right) \tan \left(\frac{x}{2} - \frac{\pi}{8} \right)$$

$$f'' \left(\frac{\pi}{12} \right) = \frac{1}{2} \left(1 + \tan^2 \frac{\pi}{12} \right) \tan \frac{\pi}{12}$$

$$= \frac{1}{2} \left(1 + (2 - \sqrt{3})^2 \right) (2 - \sqrt{3})$$

$$= \frac{1}{2} (1 + 4 + 3 - 4\sqrt{3}) (2 - \sqrt{3})$$

$$= \frac{1}{2} (8 - 4\sqrt{3}) (2 - \sqrt{3})$$

$$= (4 - 2\sqrt{3}) (2 - \sqrt{3})$$

$$= 4(2 - \sqrt{3})^2$$

$$= 4(4 + 3 - 4\sqrt{3})$$

$$= 28 - 16\sqrt{3}$$

$$f \left(\frac{\pi}{12} \right) + f'' \left(\frac{\pi}{12} \right) = 2 - \sqrt{3} + (28 - 16\sqrt{3})$$

$$= 2 - \sqrt{3} + 28 - 16\sqrt{3}$$

$$= 30 - 17\sqrt{3}$$

Question: If all the letters of the word INDEPENDENCE are arranged randomly, the find the probability that all vowels are together.

Answer: $\frac{1}{99}$

Solution:

Given word INDEPENDENCE

I E E E E

N D P N D N C

I-1 E-4 N-3 D-2 P-1 C-1

$$P = \frac{\frac{8!}{3!2!} \times \frac{5!}{4!}}{12!}$$

$$= \frac{120}{12 \times 11 \times 10 \times 9}$$

$$= \frac{1}{99}$$

Question: How many subsets of $A \times B$ are possible such that it has no more than 6 elements and atleast 3 elements if 'A' has 5 elements 'B' has 2 elements?

Answer: 792.00

Solution:

$$A \rightarrow 5$$

$$B \rightarrow 2$$

$$n(A \times B) = 10$$

$${}^{10}C_3 + {}^{10}C_4 + {}^{10}C_5 + {}^{10}C_6 = 792$$

Question: If $\left(e, \frac{4}{3}\right)$ & (e^4, α) satisfies the differential equation $\frac{dy}{dx} + \frac{y}{2x \ln x} = \frac{1}{x}$, then find α .

Answer: 3.00

Solution:

$$\frac{dy}{dx} + \frac{y}{2x \ln x} = \frac{1}{x}$$

$$\text{IF} = e^{\frac{1}{x} \int \frac{1}{x \ln x} dx}$$

$$= e^{\frac{1}{2} \ln(\ln x)}$$

$$= \sqrt{\ln x}$$

$$y \times (\sqrt{\ln x}) = \int \frac{\sqrt{\ln x}}{x} = \frac{2(\ln x)^{\frac{3}{2}}}{3} + C$$

$$\frac{4}{3} = \frac{2}{3} + C \Rightarrow C = \frac{2}{3}$$

$$(e^4, \alpha) \Rightarrow \alpha \times 2 = \frac{2}{3} \times 8 + \frac{2}{3}$$

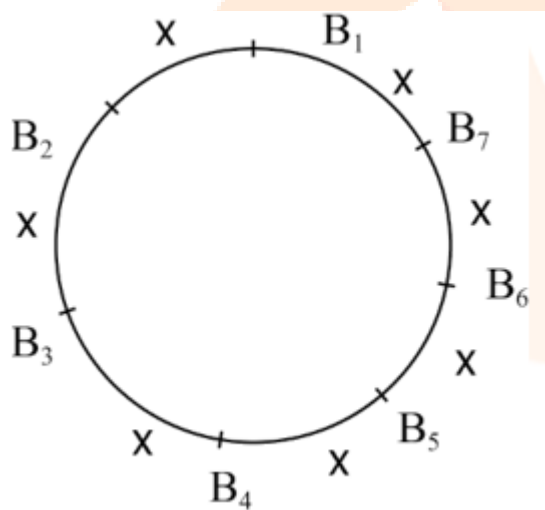
$$2\alpha = 6$$

$$\Rightarrow \alpha = 3$$

Question: How many ways are there to arrange 5 girls and 7 boys on a circular table such that no two girls are together?

Answer: $6! \times {}^7C_5 \times 5!$

Solution:



Arrangement of boys = $6!$

Arrangement of girls = ${}^7C_5 \times 5!$

The number of ways = $6! \times {}^7C_5 \times 5!$

Question: If $t_n = \frac{n^3}{n^4 + 147}$; $n \in N$. Find maximum value of t_n .

Answer: $\frac{125}{772}$

Solution:

$$f(x) = \frac{x^3}{x^4 + 147}$$

$$f'(x) = \frac{(x^4 + 147) \times 3x^2 - x^3(4x^3)}{(x^4 + 147)^2}$$

$$= \frac{x^2[3x^4 + 441 - 4x^4]}{(x^4 + 147)^2}$$

$$= \frac{x^2((21)^2 - x^4)}{(x^4 + 147)^2}$$

Maximum at $x = \sqrt{21}$

Ans is $x = 4$ or 5

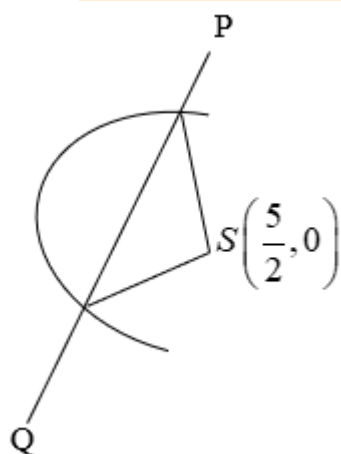
$$f(4) < f(5)$$

$$f(5) = \frac{125}{772}$$

Question: Line $y = mx + c$ intersects $y^2 = bx$ at P and Q. Focus of $y^2 = 10x$ is at S. Centroid of ΔPQS is $(10, 10)$. If $c - m = 6$, then the value of $|PQ|^2 = ?$

Answer: 1400.00

Solution:



$$y^2 = 10\left(\frac{y-c}{m}\right)$$

$$y^2 - \frac{10}{m}y + \frac{10c}{m} = 0$$

$$\frac{\frac{10}{m} + 0}{3} = 10$$

$$m = \frac{1}{3}$$

$$c = \frac{19}{3}$$

$$\text{So } y^2 - 30y + 190 = 0$$

$$|y_2 - y_1| = \sqrt{140}$$

$$|PQ|^2 = (y_2 - y_1)^2 \left(1 + \frac{1}{m^2} \right)$$

$$= 140 \times 10 = 1400$$

Question: If $S_k = \frac{1+2+\dots+k}{k}$, then $\sum_{k=1}^n (S_k)^2 = ?$

Answer: ()

Solution:

$$S_k = \frac{1+2+\dots+k}{k}$$

$$\sum_{k=1}^n (S_k)^2 = \sum_{k=1}^n \left(\frac{k+1}{2} \right)^2 = \frac{1}{4} \left[\frac{(n+1)(n+2)(2n+3)}{6} - 1 \right]$$

Question: $\lim_{x \rightarrow 0} \frac{1 - \cos^2 3x}{\cos^3 4x} \times \frac{\sin^3 4x}{[\ln(1+2x)]^5} =$

Answer: 18.00

Solution:

$$\lim_{x \rightarrow 0} \frac{1 - \cos^2 3x}{\cos^3 4x} \times \frac{\sin^3 4x}{[\ln(1+2x)]^5}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos^2 3x}{9x^2} \times \frac{\sin^3 4x}{64x^3} \times \frac{32x^5}{[\ln(1+2x)]^5} \times \frac{18}{\cos^3 4x}$$

$$= 2 \times \frac{1}{2} \times 18$$

$$= 18$$

Question: Find the coefficient of x^0 in the expansion of $\left(3x^2 - \frac{1}{2x^5}\right)^7$.

Answer: ()

Solution:

$$\begin{aligned}\text{Coefficient} &= {}^7C_2 (3x^2)^5 \times \left(-\frac{1}{2x^5}\right)^2 \\ &= \frac{{}^7C_2 \times 3^5}{2^2} \\ &= \frac{5103}{4}\end{aligned}$$

Question: A bolt manufacturing factory has three products A, B & C. 50% and 30% of the product are A and B type respectively and remaining are C type. Then probability that the product A is defective is 4%, that of B is 3% and that of C is 2%. A product is picked randomly and found to be defective, then the probability that it is of type C is:

Answer: $\frac{4}{33}$

Solution:

$$P(A) = 0.5$$

$$P(B) = 0.3$$

$$P(C) = 0.2$$

$$P\left(\frac{D}{A}\right) = 0.04$$

$$P\left(\frac{D}{B}\right) = 0.03$$

$$P\left(\frac{D}{C}\right) = 0.02$$

$$\left(\frac{C}{D}\right) = \frac{P(C) \times P\left(\frac{D}{C}\right)}{\sum P(A) \times P\left(\frac{D}{A}\right)}$$

Substituting values we get

$$\frac{4}{33}$$

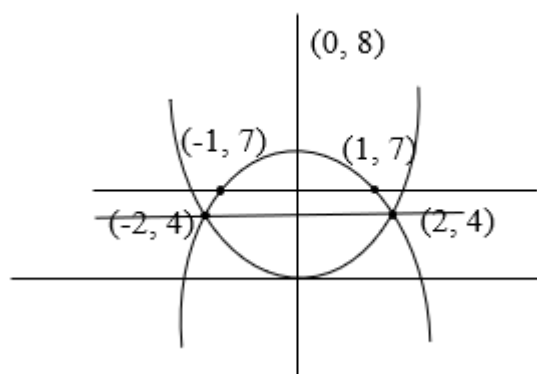
Question: The area under the curves $x^2 \leq y$, $y \leq 8 - x^2$ and $y \leq 7$ is _____.

Answer: 20.00

Solution:

$$x^2 = y$$

$$x^2 = -(y-8)$$



$$\frac{64}{6} + \frac{64}{6} - \frac{8}{6} = 20$$

Question: $\int \frac{x+1}{x(1+xe^x)^2} dx$

Answer:

Solution:

$$I = \int \frac{x+1}{x(1+xe^x)^2} dx$$

$$\frac{d}{dx}(xe^x) = e^x(x+1)$$

$$(1+xe^x) = t$$

$$e^x(1+x)dx = dt$$

$$I = \int \frac{dt}{(t-1)t^2}$$

$$= \int \frac{dt}{t^2(t-1)}$$

$$\left[\frac{1}{t} \times \left(\frac{1}{t(t-1)} \right) \right] = \frac{1}{t} \left(\frac{1}{t-1} - \frac{1}{t} \right)$$

$$= \frac{1}{t(t-1)} - \frac{1}{t^2}$$

$$= \frac{1}{t-1} - \frac{1}{t} - \frac{1}{t^2}$$

$$I = \ln|t-1| - \ln|t| + \frac{1}{t} + C$$

$$= \ln|xe^x| - \ln|1+xe^x| + \frac{1}{1+xe^x} + C$$

$$I(x) = \ln \left| \frac{xe^x}{1+xe^x} \right| + \frac{1}{1+xe^x} + 1$$

$$\therefore I(1) = \ln \left| \frac{e}{1+e} \right| + \frac{1}{1+e} + 1$$

$$= 2 - \ln(1+e) + \frac{1}{1+e}$$

Question: $\sim((q \rightarrow p) \rightarrow (p \rightarrow q))$ is equivalent to

Options:

- (a)
- (b)
- (c)
- (d)

Answer: (d)

Solution:

$$\equiv \sim(\sim(q \rightarrow p) \vee (p \rightarrow q))$$

$$\equiv (q \rightarrow p) \wedge (p \rightarrow q)$$

$$\equiv (\sim q \vee p) \wedge (\sim p \vee q)$$

$$\equiv (\sim q \vee p) \wedge (p \wedge \sim q)$$

$$\equiv p \wedge \sim q$$

Question: If α, β, γ are the roots of the equation $x^3 + cx + d = 0$ and $\gamma\beta = 1 = -\alpha$, then find relation which b satisfies.

Answer: $b = 0$

Solution:

$$\alpha = -1, \alpha\beta = 1$$

$$\alpha + \beta + \gamma = 0$$

$$\beta + \alpha = 1$$

$$x^2 - x + 1 = 0$$

$$\text{Now, } S_2 = b$$

$$-\beta + \beta\gamma + (-\alpha) = b$$

$$b = -1 + 1$$

$$b = 0$$

